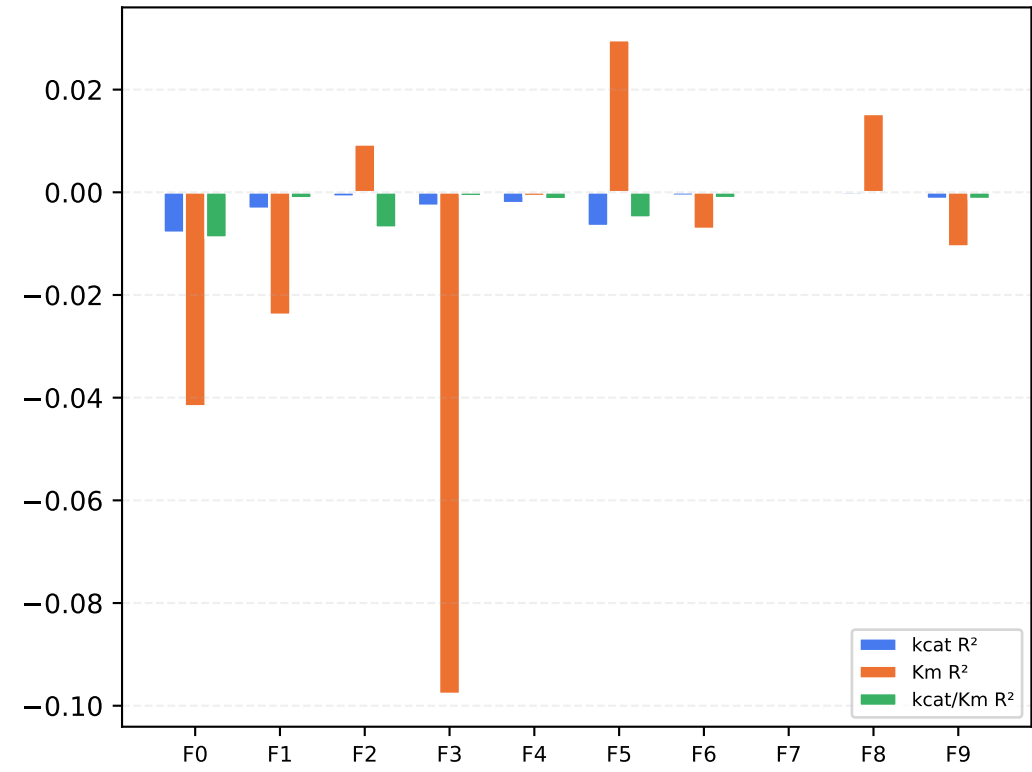
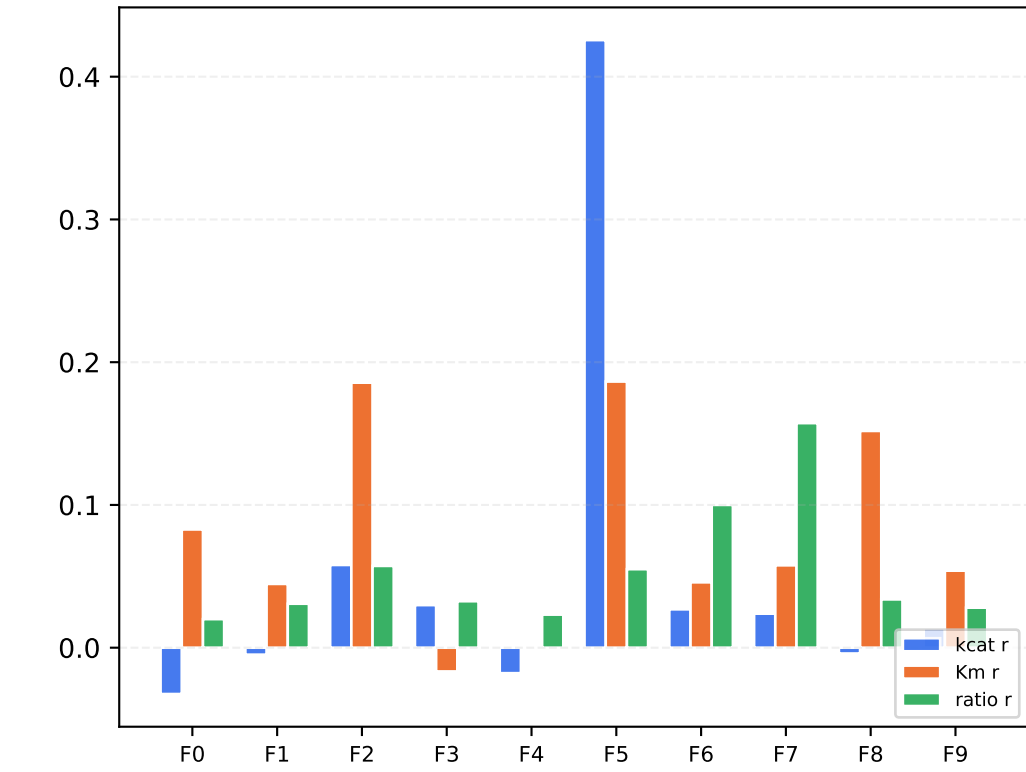


10-Fold Cross-Validation: Comprehensive Report

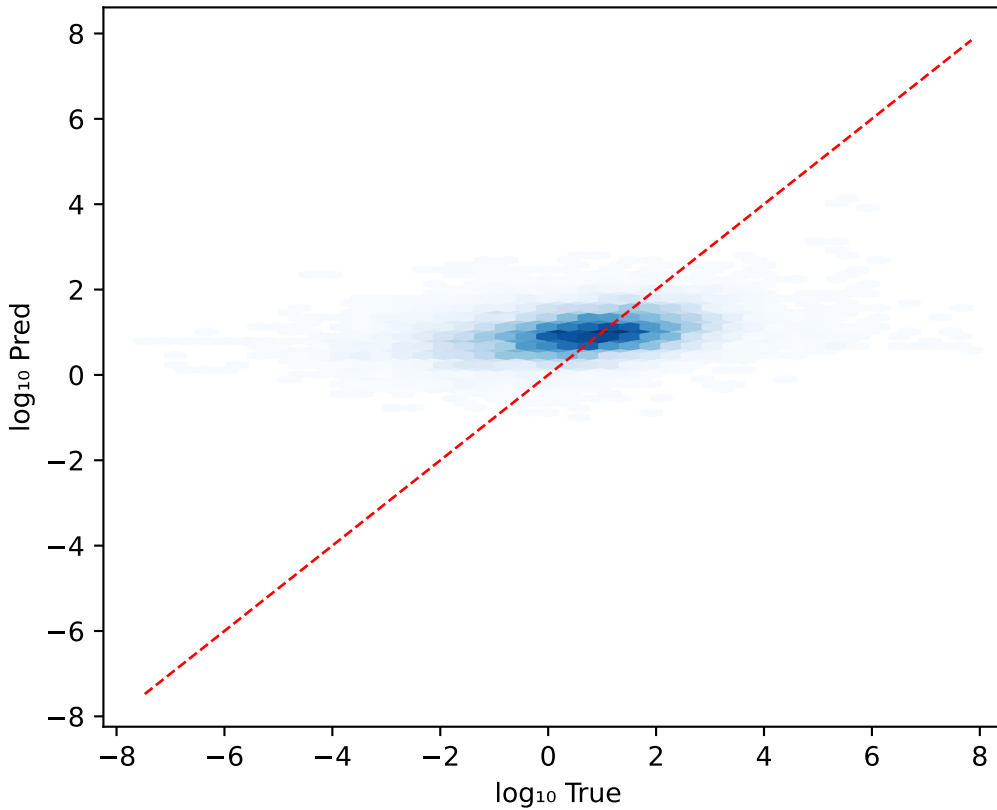
R² per Fold



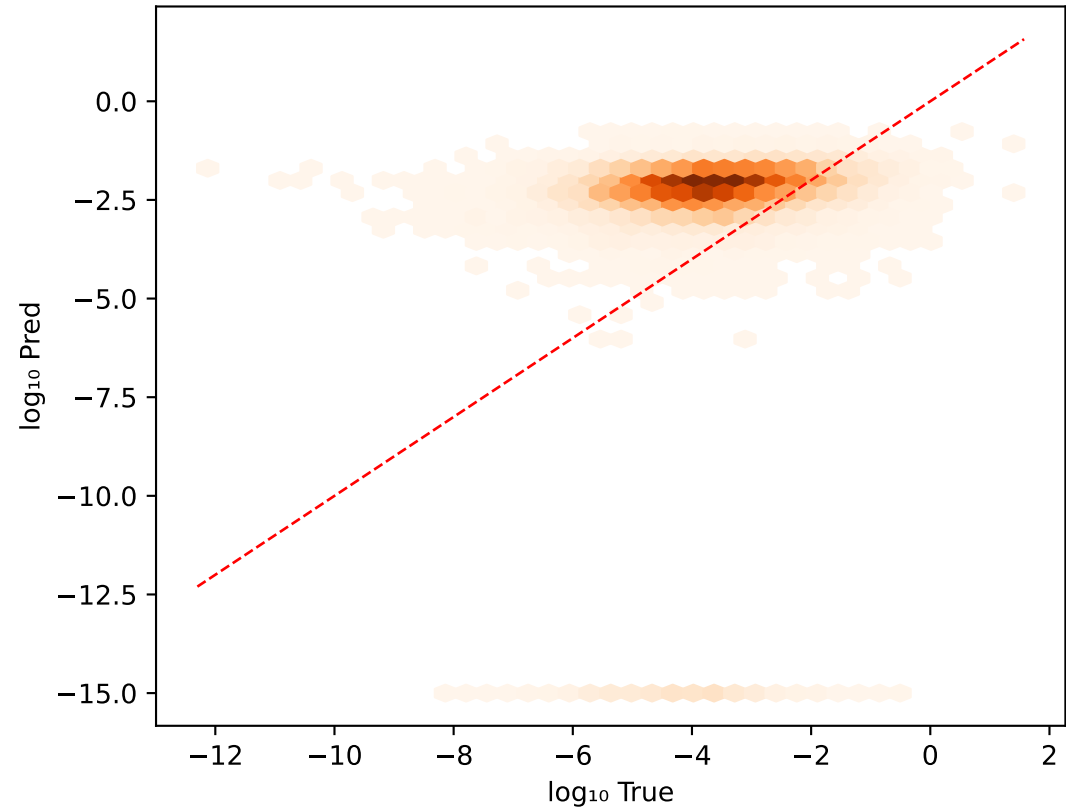
Pearson r per Fold



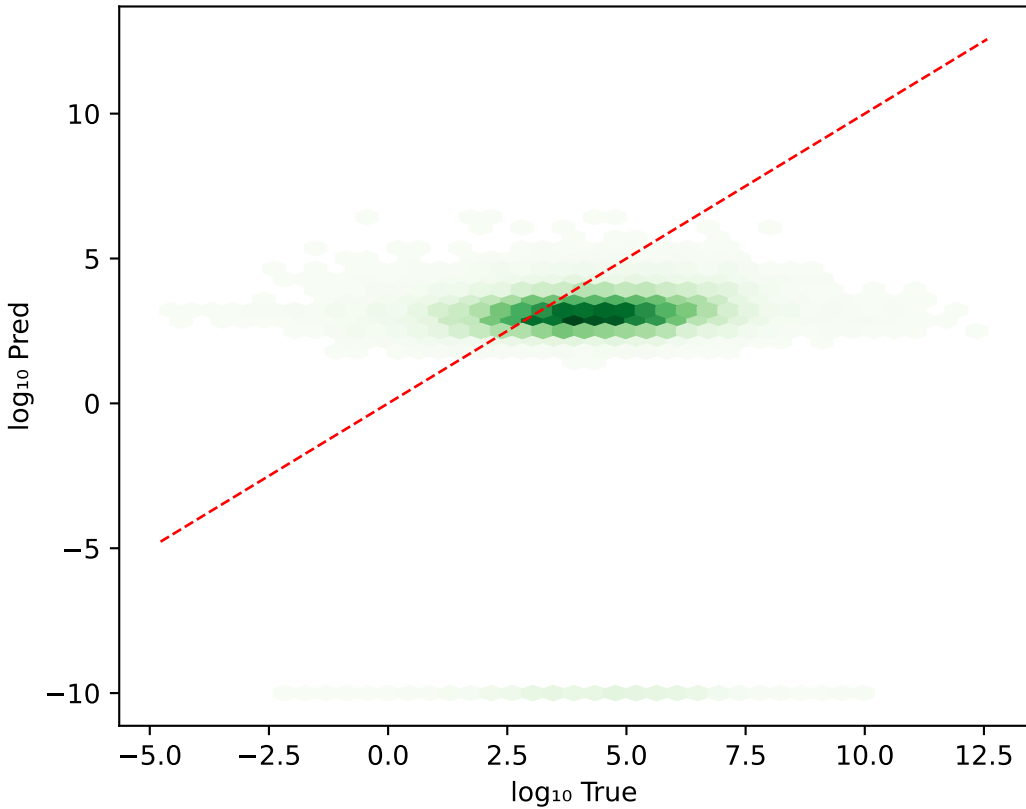
kcat: Pred vs Exp
R²=-0.000



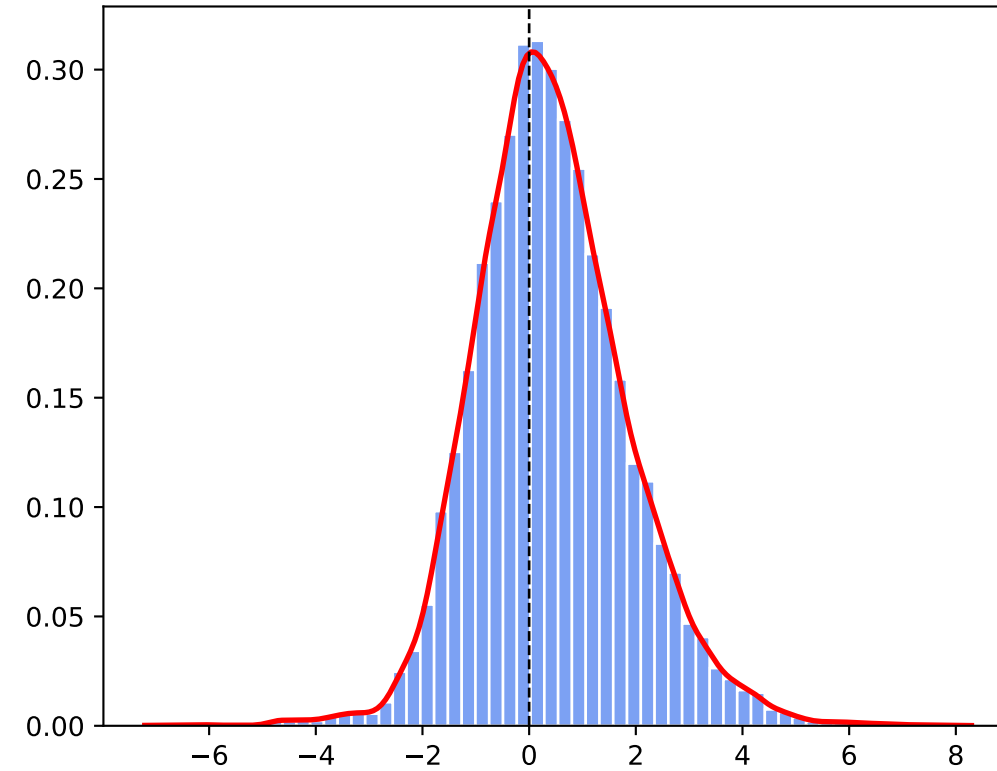
Km: Pred vs Exp
R²=0.001



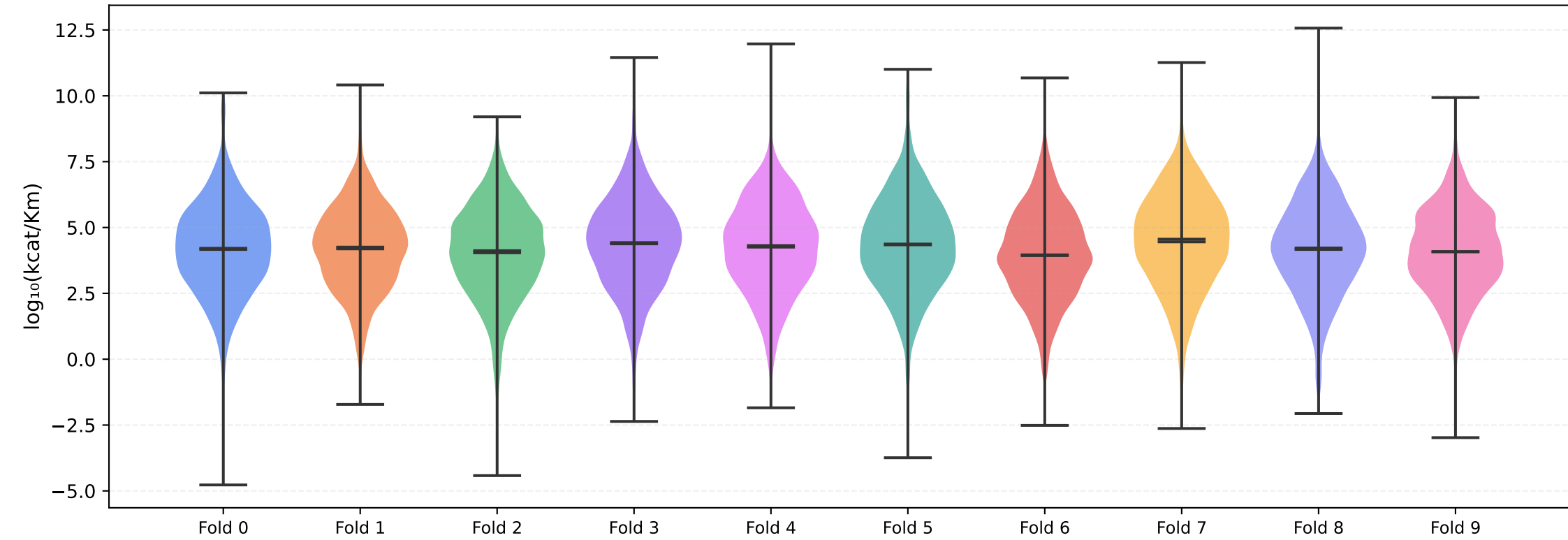
kcat/Km: Pred vs Exp
R²=-0.000



kcat Residuals
μ=0.421 σ=1.430



kcat/Km Distribution by Fold



10-Fold CV Summary

kcat:
R² = -0.0026 ± 0.0025
r = 0.0518 ± 0.1271
MAE = 1.54e+04

Km:
R² = -0.0128 ± 0.0340
r = 0.0793 ± 0.0685

kcat/Km:
R² = -0.0027 ± 0.0029
r = 0.0538 ± 0.0412

Model: LightGBM (100 trees, max_depth=10)
Features: ESMC 300M (960-dim)
Folds: 10 (protein-aware)
Samples: 25,831